Machine Learning Concepts

Sample knowledge-base document for a Retrieval-Augmented Generation project.

Supervised Learning

Supervised learning trains a model using input-output pairs. In classification, the output is a category. In regression, the output is a continuous value. Examples include spam detection, image classification, house price prediction, and demand forecasting.

Train, Validation, and Test Sets

A training set is used to fit model parameters. A validation set is used for model selection and hyperparameter tuning. A test set is reserved for final evaluation. Keeping these roles separate helps reduce misleading performance estimates.

Overfitting and Underfitting

Overfitting occurs when a model learns training data too specifically and performs poorly on new data. Underfitting occurs when a model is too simple to capture important patterns. Regularization, more data, early stopping, cross-validation, and appropriate model complexity can help.

Feature Engineering

Features are the input variables used by a model. Feature engineering may include scaling numeric values, encoding categories, handling missing values, creating interaction terms, or deriving domain-specific variables.

Common Algorithms

Linear regression models continuous outcomes. Logistic regression is commonly used for classification. Decision trees split data using feature-based rules. Random forests combine many trees to improve robustness. Gradient boosting builds trees sequentially to correct prior errors.

Model Metrics

Accuracy is useful when classes are balanced. Precision measures how many predicted positives are correct. Recall measures how many actual positives are found. F1 score balances precision and recall. Confusion matrices show the counts of correct and incorrect class predictions.